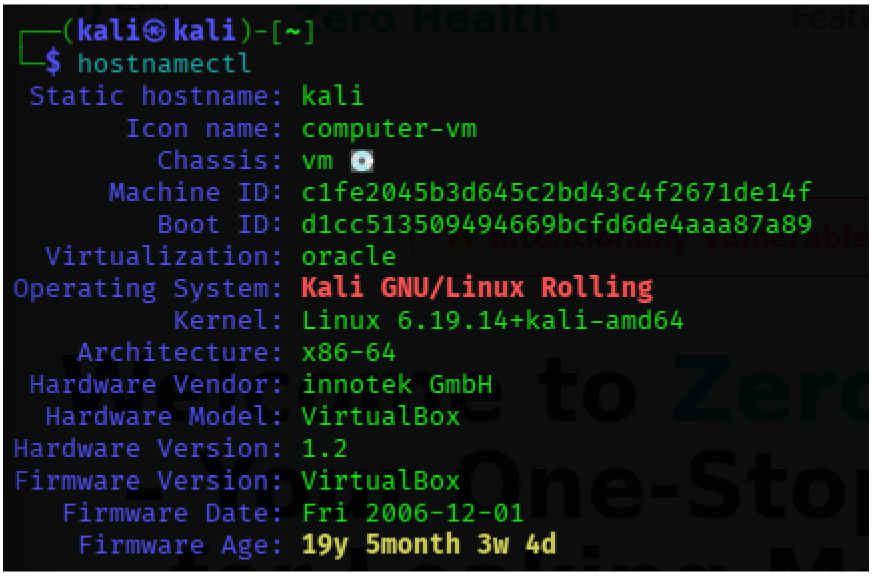
Lab 1:Introduction to Kali Linux

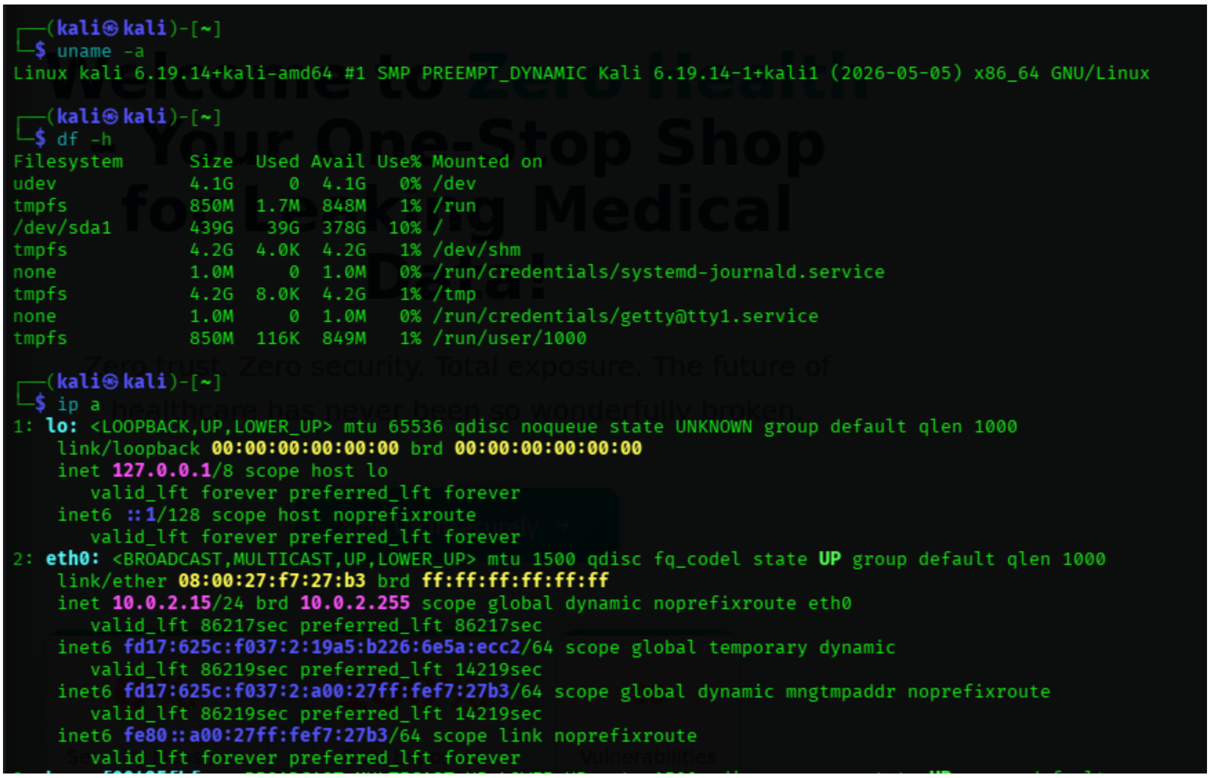
Purpose

This lab introduces the Kali Linux environment and examines its role in cybersecurity workflows. Kali Linux is a specialized distribution for penetration testing, security assessments, and forensic analysis.

Basic command line

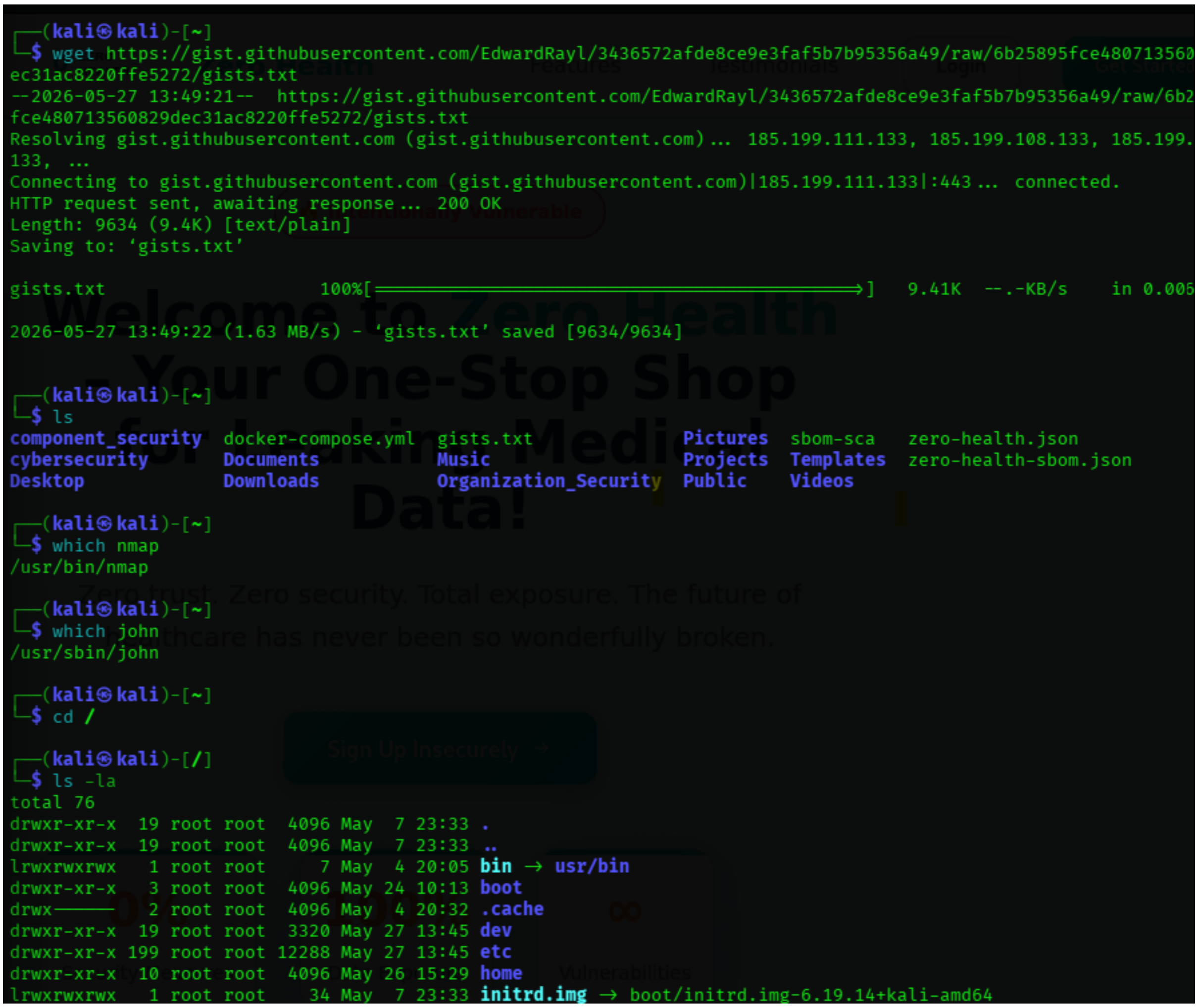
The screenshots demonstrate common system-verification commands, including whoami, hostnamectl, uname -a, df -h, and ip.



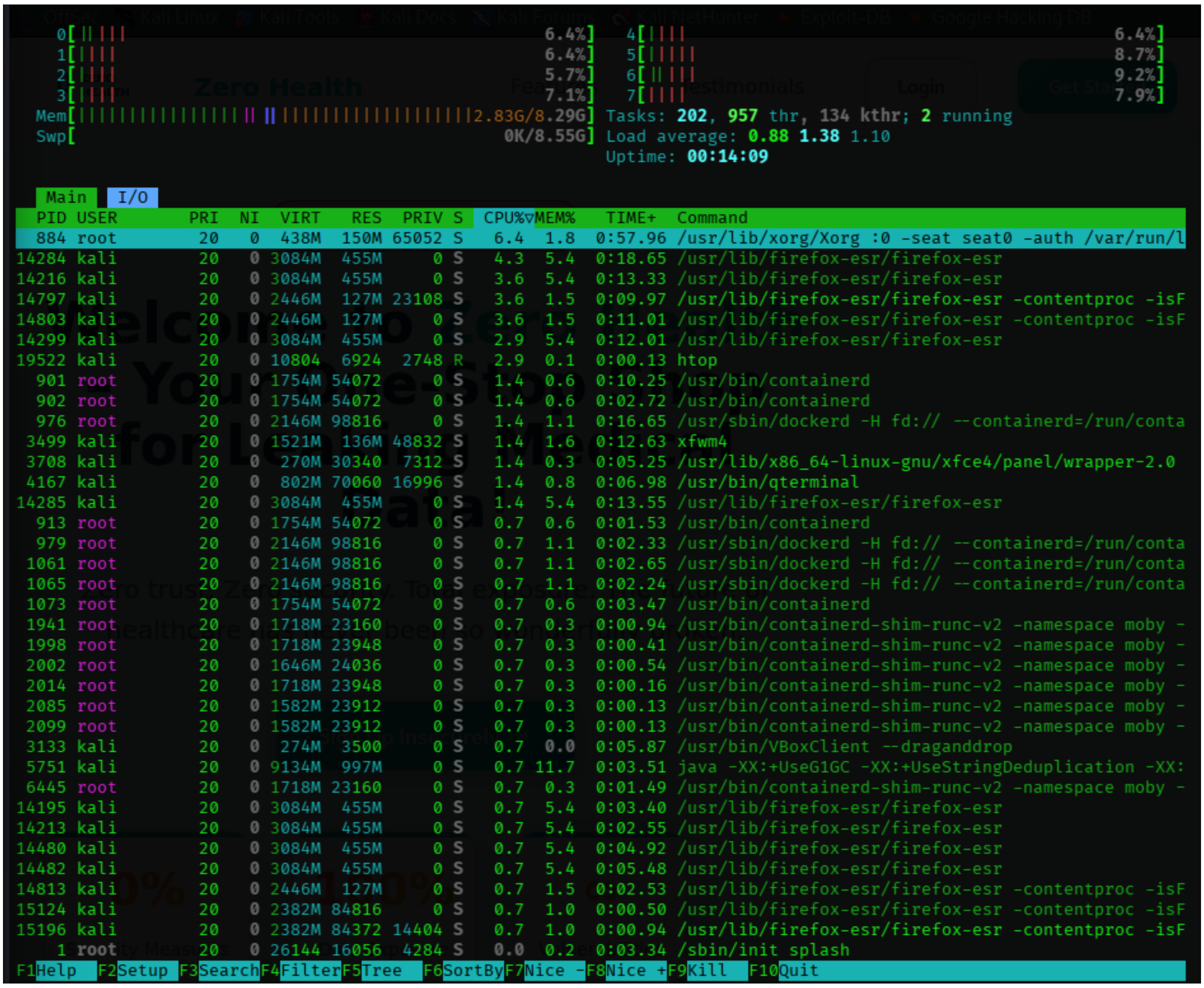


```
┌──(kali㉿kali)-[~]
└─$ uname -a
Linux kali 6.19.14+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.19.14-1+kali1 (2026-05-05) x86_64 GNU/Linux

┌──(kali㉿kali)-[~]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            4.1G     0  4.1G   0% /dev
tmpfs           850M  1.7M  848M   1% /run
/dev/sda1       439G   39G  378G  10% /
tmpfs           4.2G  4.0K  4.2G   1% /dev/shm
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           4.2G  8.0K  4.2G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           850M  116K  849M   1% /run/user/1000

┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f7:27:b3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86217sec preferred_lft 86217sec
    inet6 fd17:625c:f037:2:19a5:b226:6e5a:ecc2/64 scope global temporary dynamic
       valid_lft 86219sec preferred_lft 14219sec
    inet6 fd17:625c:f037:2:a00:27ff:fef7:27b3/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86219sec preferred_lft 14219sec
    inet6 fe80::a00:27ff:fef7:27b3/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

The screenshots also demonstrate using tools such as wget, nmap, john, htop, traceroute, nload, strings, tcpdump, searchsploit, dnsenum, lbd, and nth.

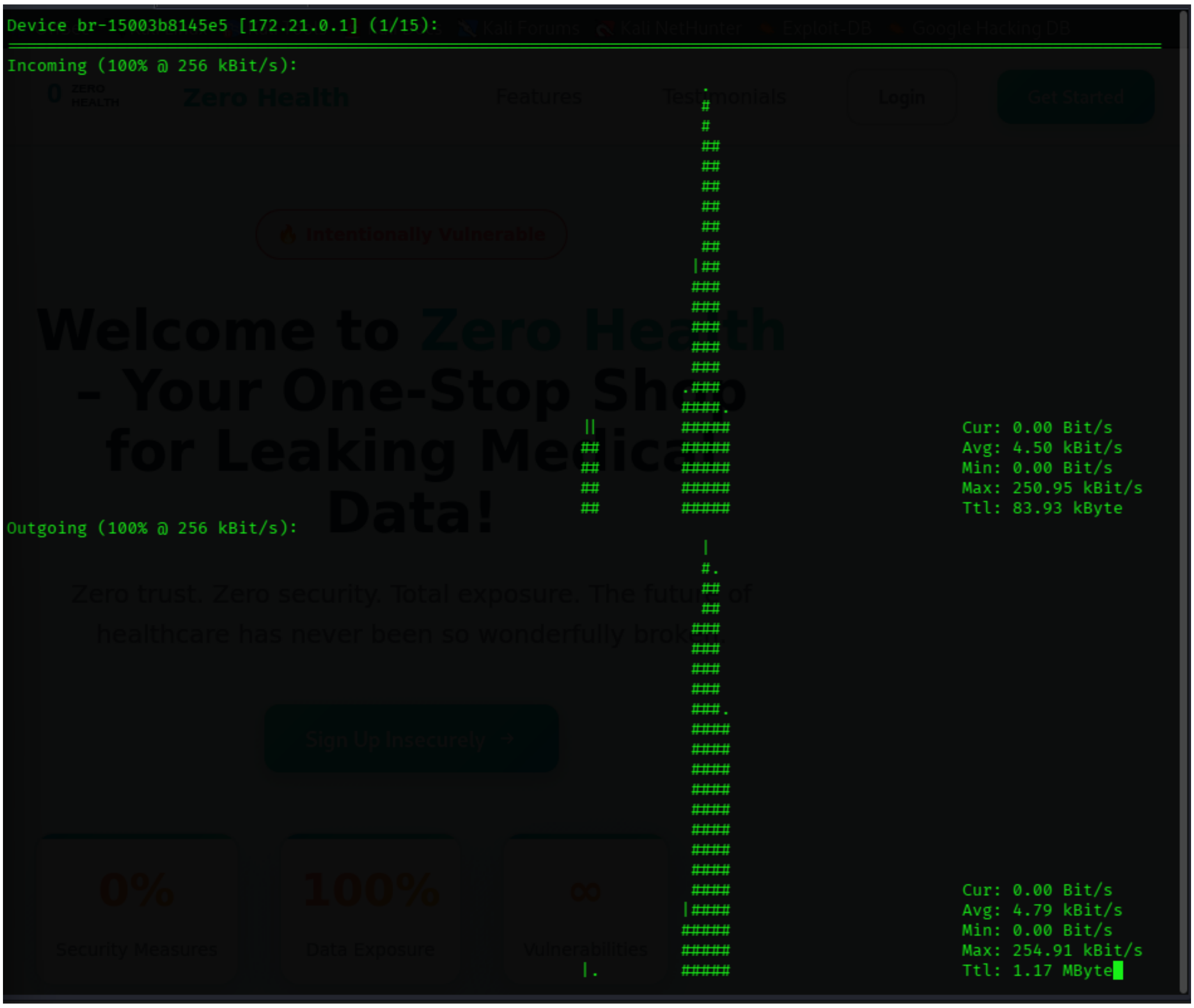


```
┌──(kali㉿kali)-[~]
└─$ wget https://gist.githubusercontent.com/EdwardRayl/3436572afde8ce9e3faf5b7b95356a49/raw/6b25895fce48071356
ec31ac8220ffe5272/gists.txt
--2026-05-27 13:49:21--  https://gist.githubusercontent.com/EdwardRayl/3436572afde8ce9e3faf5b7b95356a49/raw/6b2
fce480713560829dec31ac8220ffe5272/gists.txt
Resolving gist.githubusercontent.com (gist.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.
133, ...
Connecting to gist.githubusercontent.com (gist.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9634 (9.4K) [text/plain]
Saving to: ‘gists.txt’

gists.txt                 100%[=====================================>]   9.41K  --.-KB/s    in 0.005

2026-05-27 13:49:22 (1.63 MB/s) - ‘gists.txt’ saved [9634/9634]


┌──(kali㉿kali)-[~]
└─$ ls
component_security  docker-compose.yml  gists.txt             Pictures  sbom-sca   zero-health.json
cybersecurity       Documents           Music                 Projects  Templates  zero-health-sbom.json
Desktop             Downloads           Organization_Security Public    Videos

┌──(kali㉿kali)-[~]
└─$ which nmap
/usr/bin/nmap

┌──(kali㉿kali)-[~]
└─$ which john
/usr/sbin/john

┌──(kali㉿kali)-[~]
└─$ cd /

┌──(kali㉿kali)-[/]
└─$ ls -la
total 76
drwxr-xr-x  19 root root  4096 May  7 23:33 .
drwxr-xr-x  19 root root  4096 May  7 23:33 ..
lrwxrwxrwx   1 root root     7 May  4 20:05 bin -> usr/bin
drwxr-xr-x   3 root root  4096 May 24 10:13 boot
drwx------   2 root root  4096 May  4 20:32 .cache
drwxr-xr-x  19 root root  3320 May 27 13:45 dev
drwxr-xr-x 199 root root 12288 May 27 13:45 etc
drwxr-xr-x  10 root root  4096 May 26 15:29 home
lrwxrwxrwx   1 root root    34 May  7 23:33 initrd.img -> boot/initrd.img-6.19.14+kali-amd64
```



0[||||| 6.4%]
1[|||| 6.4%]
2[|||| 5.7%]
3[|||| 7.1%]
4[|||| 6.4%]
5[||||| 8.7%]
6[||||| 9.2%]
7[|||| 7.9%]
Mem[||||||||||||||||||||||||||||||||||||2.83G/8.29G]
Swp[0K/8.55G]
Tasks: 202, 957 thr, 134 kthr; 2 running
Load average: 0.88 1.38 1.10
Uptime: 00:14:09
Main
I/O
PID USER PRI NI VIRT RES PRIV S CPU%▽MEM% TIME+ Command
884 root 20 0 438M 150M 65052 S 6.4 1.8 0:57.96 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/l
14284 kali 20 0 3084M 455M 0 S 4.3 5.4 0:18.65 /usr/lib/firefox-esr/firefox-esr
14216 kali 20 0 3084M 455M 0 S 3.6 5.4 0:13.33 /usr/lib/firefox-esr/firefox-esr
14797 kali 20 0 2446M 127M 23108 S 3.6 1.5 0:09.97 /usr/lib/firefox-esr/firefox-esr -contentproc -isF
14803 kali 20 0 2446M 127M 0 S 3.6 1.5 0:11.01 /usr/lib/firefox-esr/firefox-esr -contentproc -isF
14299 kali 20 0 3084M 455M 0 S 2.9 5.4 0:12.01 /usr/lib/firefox-esr/firefox-esr
19522 kali 20 0 10804 6924 2748 R 2.9 0.1 0:00.13 htop
901 root 20 0 1754M 54072 0 S 1.4 0.6 0:10.25 /usr/bin/containerd
902 root 20 0 1754M 54072 0 S 1.4 0.6 0:02.72 /usr/bin/containerd
976 root 20 0 2146M 98816 0 S 1.4 1.1 0:16.65 /usr/sbin/dockerd -H fd:// --containerd=/run/conta
3499 kali 20 0 1521M 136M 48832 S 1.4 1.6 0:12.63 xfwm4
3708 kali 20 0 270M 30340 7312 S 1.4 0.3 0:05.25 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0
4167 kali 20 0 802M 70060 16996 S 1.4 0.8 0:06.98 /usr/bin/qterminal
14285 kali 20 0 3084M 455M 0 S 1.4 5.4 0:13.55 /usr/lib/firefox-esr/firefox-esr
913 root 20 0 1754M 54072 0 S 0.7 0.6 0:01.53 /usr/bin/containerd
979 root 20 0 2146M 98816 0 S 0.7 1.1 0:02.33 /usr/sbin/dockerd -H fd:// --containerd=/run/conta
1061 root 20 0 2146M 98816 0 S 0.7 1.1 0:02.65 /usr/sbin/dockerd -H fd:// --containerd=/run/conta
1065 root 20 0 2146M 98816 0 S 0.7 1.1 0:02.24 /usr/sbin/dockerd -H fd:// --containerd=/run/conta
1073 root 20 0 1754M 54072 0 S 0.7 0.6 0:03.47 /usr/bin/containerd
1941 root 20 0 1718M 23160 0 S 0.7 0.3 0:00.94 /usr/bin/containerd-shim-runc-v2 -namespace moby -
1998 root 20 0 1718M 23948 0 S 0.7 0.3 0:00.41 /usr/bin/containerd-shim-runc-v2 -namespace moby -
2002 root 20 0 1646M 24036 0 S 0.7 0.3 0:00.54 /usr/bin/containerd-shim-runc-v2 -namespace moby -
2014 root 20 0 1718M 23948 0 S 0.7 0.3 0:00.16 /usr/bin/containerd-shim-runc-v2 -namespace moby -
2085 root 20 0 1582M 23912 0 S 0.7 0.3 0:00.13 /usr/bin/containerd-shim-runc-v2 -namespace moby -
2099 root 20 0 1582M 23912 0 S 0.7 0.3 0:00.13 /usr/bin/containerd-shim-runc-v2 -namespace moby -
3133 kali 20 0 274M 3500 0 S 0.7 0.0 0:05.87 /usr/bin/VBoxClient --draganddrop
5751 kali 20 0 9134M 997M 0 S 0.7 11.7 0:03.51 java -XX:+UseG1GC -XX:+UseStringDeduplication -XX:
6445 root 20 0 1718M 23160 0 S 0.7 0.3 0:01.49 /usr/bin/containerd-shim-runc-v2 -namespace moby -
14195 kali 20 0 3084M 455M 0 S 0.7 5.4 0:03.40 /usr/lib/firefox-esr/firefox-esr
14213 kali 20 0 3084M 455M 0 S 0.7 5.4 0:02.55 /usr/lib/firefox-esr/firefox-esr
14480 kali 20 0 3084M 455M 0 S 0.7 5.4 0:04.92 /usr/lib/firefox-esr/firefox-esr
14482 kali 20 0 3084M 455M 0 S 0.7 5.4 0:05.48 /usr/lib/firefox-esr/firefox-esr
14813 kali 20 0 2446M 127M 0 S 0.7 1.5 0:02.53 /usr/lib/firefox-esr/firefox-esr -contentproc -isF
15124 kali 20 0 2382M 84816 0 S 0.7 1.0 0:00.50 /usr/lib/firefox-esr/firefox-esr -contentproc -isF
15196 kali 20 0 2382M 84372 14404 S 0.7 1.0 0:00.94 /usr/lib/firefox-esr/firefox-esr -contentproc -isF
1 root 20 0 26144 16056 4284 S 0.0 0.2 0:03.34 /sbin/init splash
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice - F8Nice + F9Kill F10Quit

```
┌──(kali㉿kali)-[/]
└─$ traceroute google.com
traceroute to google.com (142.251.223.142), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  4.191 ms  3.868 ms  3.793 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```




```
┌──(kali㉿kali)-[/]
└─$ strings /bin/ls | head
t/lib64/ld-linux-x86-64.so.2
_ITM_deregisterTMCloneTable
__gmon_start__
_ITM_registerTMCloneTable
fgetfilecon_raw
fgetfilecon
freecon
lgetfilecon
lgetfilecon_raw
optind
```

```
┌──(kali㉿kali)-[/]
└─$ sudo tcpdump -c 10
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
15:29:14.394126 IP kali.56879 > reliance.reliance.domain: 26948+ AAAA? db. (20)
15:29:14.394738 IP kali.42519 > reliance.reliance.domain: 26798+ A? db. (20)
15:29:14.397889 IP reliance.reliance.domain > kali.56879: 26948 NXDomain 0/0/0 (20)
15:29:14.400122 IP reliance.reliance.domain > kali.42519: 26798 NXDomain 0/0/0 (20)
15:29:14.432887 IP kali.33273 > reliance.reliance.domain: 11359+ PTR? 1.29.168.192.in-addr.arpa. (43)
15:29:14.446301 IP reliance.reliance.domain > kali.33273: 11359* 1/0/0 PTR reliance.reliance. (74)
15:29:16.407209 IP kali.32936 > reliance.reliance.domain: 24378+ AAAA? db. (20)
15:29:16.407887 IP kali.41310 > reliance.reliance.domain: 24048+ A? db. (20)
15:29:16.416620 IP reliance.reliance.domain > kali.32936: 24378 NXDomain 0/0/0 (20)
15:29:16.416621 IP reliance.reliance.domain > kali.41310: 24048 NXDomain 0/0/0 (20)
10 packets captured
10 packets received by filter
0 packets dropped by kernel
```

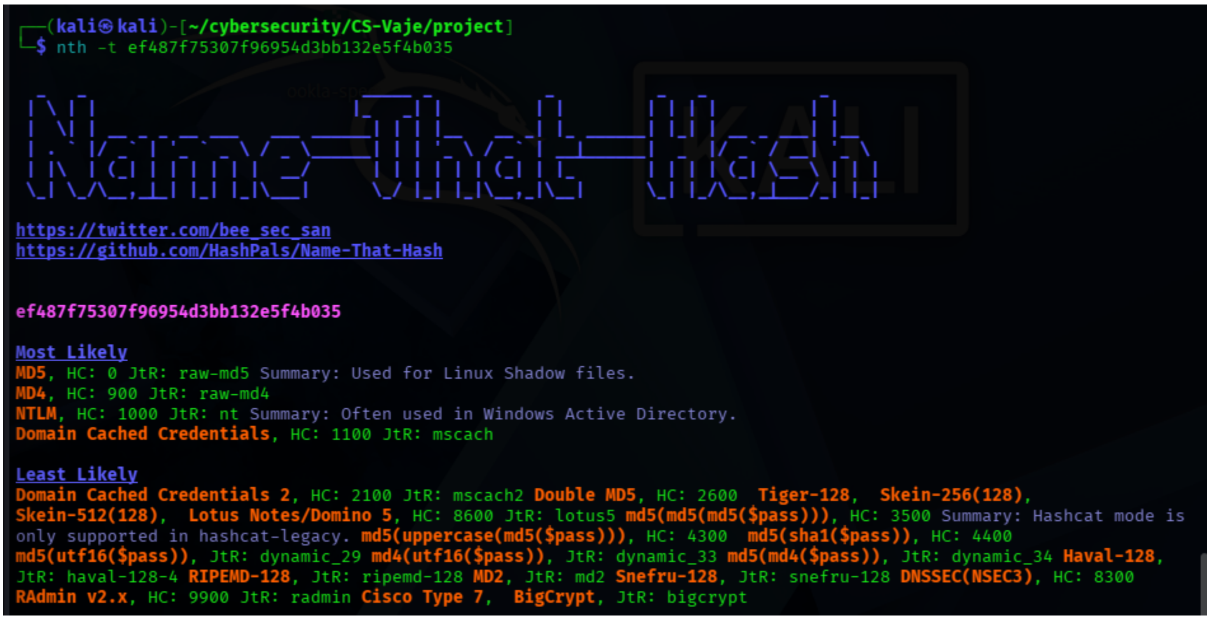
```
┌──(kali㉿kali)-[~/cybersecurity/CS-Vaje/project]
└─$ searchsploit wordpress ftp
----------------------------------------------------------------------------------- ---------------------------------
 Exploit Title                                                                     |  Path
----------------------------------------------------------------------------------- ---------------------------------
WordPress Plugin MiwoFTP 1.0.5 - Arbitrary File Download (1)                       | php/webapps/36774.txt
WordPress Plugin MiwoFTP 1.0.5 - Arbitrary File Download (2)                       | php/webapps/36801.txt
WordPress Plugin MiwoFTP 1.0.5 - Cross-Site Request Forgery / Arbitrary File Cre   | php/webapps/36763.txt
WordPress Plugin MiwoFTP 1.0.5 - Cross-Site Request Forgery / Arbitrary File Del   | php/webapps/36761.txt
WordPress Plugin MiwoFTP 1.0.5 - Multiple Cross-Site Request Forgery / Cross-Sit   | php/webapps/36762.txt
----------------------------------------------------------------------------------- ---------------------------------
Shellcodes: No Results
```

```
┌──(kali㉿kali)-[~/cybersecurity/CS-Vaje/project]
└─$ dnsenum google.com
dnsenum VERSION:1.3.1

-----   google.com   -----

Host's addresses:
__________________

google.com.                              249      IN    A        142.251.223.142

Name Servers:
______________

ns4.google.com.                          89475    IN    A        216.239.38.10
ns2.google.com.                          157263   IN    A        216.239.34.10
ns1.google.com.                          157262   IN    A        216.239.32.10
ns3.google.com.                          135211   IN    A        216.239.36.10

Mail (MX) Servers:
___________________

smtp.google.com.                         69       IN    A        192.178.211.27
smtp.google.com.                         69       IN    A        192.178.211.26

Trying Zone Transfers and getting Bind Versions:
_________________________________________________


Trying Zone Transfer for google.com on ns3.google.com ...
AXFR record query failed: corrupt packet

Trying Zone Transfer for google.com on ns1.google.com ...
AXFR record query failed: corrupt packet

Trying Zone Transfer for google.com on ns2.google.com ...
AXFR record query failed: corrupt packet

Trying Zone Transfer for google.com on ns4.google.com ...
AXFR record query failed: corrupt packet
```

```
Brute forcing with /usr/share/dnsenum/dns.txt:
_____________________________________________

about.google.com.                 300      IN    CNAME    www3.l.google.com.
www3.l.google.com.                293      IN    A        142.250.193.142
accounts.google.com.              236      IN    A        192.178.211.84
admin.google.com.                 202      IN    A        142.250.134.100
admin.google.com.                 202      IN    A        142.250.134.138
admin.google.com.                 202      IN    A        142.250.134.139
admin.google.com.                 202      IN    A        142.250.134.101
admin.google.com.                 202      IN    A        142.250.134.102
admin.google.com.                 202      IN    A        142.250.134.113
ads.google.com.                   203      IN    A        142.250.192.206
america.google.com.               300      IN    CNAME    www3.l.google.com.
www3.l.google.com.                289      IN    A        142.250.193.142
ap.google.com.                    300      IN    CNAME    www2.l.google.com.
www2.l.google.com.                219      IN    A        192.178.174.103
www2.l.google.com.                219      IN    A        192.178.174.105
www2.l.google.com.                219      IN    A        192.178.174.99
www2.l.google.com.                219      IN    A        192.178.174.147
www2.l.google.com.                219      IN    A        192.178.174.104
www2.l.google.com.                219      IN    A        192.178.174.106
apps.google.com.                  68       IN    CNAME    www3.l.google.com.
www3.l.google.com.                288      IN    A        142.250.193.142
archive.google.com.               300      IN    A        192.178.173.100
archive.google.com.               300      IN    A        192.178.173.139
archive.google.com.               300      IN    A        192.178.173.138
archive.google.com.               300      IN    A        192.178.173.113
archive.google.com.               300      IN    A        192.178.173.101
archive.google.com.               300      IN    A        192.178.173.102
asia.google.com.                  300      IN    A        192.178.193.104
asia.google.com.                  300      IN    A        192.178.193.105
asia.google.com.                  300      IN    A        192.178.193.103
asia.google.com.                  300      IN    A        192.178.193.147
asia.google.com.                  300      IN    A        192.178.193.106
asia.google.com.                  300      IN    A        192.178.193.99
blog.google.com.                  300      IN    CNAME    www.blogger.com.
www.blogger.com.                  158      IN    CNAME    blogger.l.google.com.
blogger.l.google.com.             77       IN    A        192.178.173.191
channel.google.com.               300      IN    A        172.217.26.110
d.google.com.                     300      IN    CNAME    www3.l.google.com.
www3.l.google.com.                297      IN    A        142.250.67.78
directory.google.com.             300      IN    CNAME    www3.l.google.com.
www3.l.google.com.                287      IN    A        142.250.193.142
dns.google.com.                   319      IN    A           8.8.8.8
dns.google.com.                   319      IN    A           8.8.4.4
```

```
google.com class C netranges:
______________________________

 8.8.4.0/24
 8.8.8.0/24
 64.9.224.0/24
 74.125.24.0/24
 142.250.67.0/24
 142.250.134.0/24
 142.250.146.0/24
 142.250.182.0/24
 142.250.192.0/24
 142.250.193.0/24
 142.251.43.0/24
 142.251.150.0/24
 142.251.151.0/24
 142.251.152.0/24
 142.251.153.0/24
 142.251.154.0/24
 142.251.155.0/24
 142.251.156.0/24
 142.251.157.0/24
 142.251.223.0/24
 172.217.24.0/24
 172.217.26.0/24
 192.178.173.0/24
 192.178.174.0/24
 192.178.193.0/24
 192.178.211.0/24
 216.239.32.0/24
 216.239.34.0/24
 216.239.36.0/24
 216.239.38.0/24


Performing reverse lookup on 7680 ip addresses:
________________________________________________

68.224.9.64.in-addr.arpa.          86400    IN    PTR      vpn.google.com.
69.224.9.64.in-addr.arpa.          86400    IN    PTR      vpn.google.com.
70.224.9.64.in-addr.arpa.          86400    IN    PTR      vpn.google.com.
```

```
┌──(kali㉿kali)-[~/cybersecurity/CS-Vaje/project]
└─$ lbd google.com

lbd - load balancing detector 0.4 - Checks if a given domain uses load-balancing.
                                    Written by Stefan Behte (http://ge.mine.nu)
                                    Proof-of-concept! Might give false positives.

Checking for DNS-Loadbalancing: NOT FOUND
Checking for HTTP-Loadbalancing [Server]:
 gws
 NOT FOUND

Checking for HTTP-Loadbalancing [Date]: 11:17:29, 11:17:30, 11:17:30, 11:17:30, 11:17:31, 11:17:31, 11:17:32, 11:17
:32, 11:17:33, 11:17:33, 11:17:34, 11:17:34, 11:17:35, 11:17:35, 11:17:36, 11:17:36, 11:17:36, 11:17:37, 11:17:37,
11:17:38, 11:17:38, 11:17:39, 11:17:39, 11:17:40, 11:17:40, 11:17:40, 11:17:41, 11:17:42, 11:17:43, 11:17:44, 11:17
:44, 11:17:45, 11:17:45, 11:17:46, 11:17:46, 11:17:47, 11:17:47, 11:17:47, 11:17:48, 11:17:48, 11:17:49, 11:17:49,
11:17:49, 11:17:50, 11:17:51, 11:17:51, 11:17:52, 11:17:52, 11:17:53, 11:17:53, NOT FOUND

Checking for HTTP-Loadbalancing [Diff]: FOUND
< Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce--VrnCmRVHESdvDRyF0yjdA'
'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/c
sp/gws/other-hp
< Expires: Fri, 26 Jun 2026 11:17:53 GMT
> Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-NwqF97zzH6oiwUIkw4aJ9Q'
'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/c
sp/gws/other-hp
> Expires: Fri, 26 Jun 2026 11:17:54 GMT

google.com does Load-balancing. Found via Methods: HTTP[Diff]
```



Conclusion

This lab increased familiarity with the Kali Linux interface and its essential tools. It highlights how a security-focused operating system supports hands-on learning, testing, and analysis in cybersecurity.